

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Third Semester B.Tech Degree Examination December 2021 (2019 scheme)

Course Code: MAT201**Course Name: Partial Differential equations and Complex analysis**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions. Each question carries 3 marks*

Marks

- 1 Find the partial differential equation by eliminating arbitrary functions f and g from $z = f(x) + g(y)$ (3)
- 2 Solve $\frac{\partial^2 z}{\partial x^2} = xy$ (3)
- 3 Write the three possible solutions of one dimensional wave equation. (3)
- 4 Write any two assumptions used in the derivation of one dimensional heat equation. (3)
- 5 Test the continuity at $z = 0$ of $f(z) = \begin{cases} \frac{\text{Im}(z)}{|z|}, & z \neq 0 \\ 0, & z = 0 \end{cases}$ (3)
- 6 Check whether $f(z) = \bar{z}$ is an analytic function? (3)
- 7 Evaluate $\oint_c \frac{e^z}{z-5} dz$, where c is the circle $|z|=4$ (3)
- 8 Find the Taylor series expansion of e^z about $z = \pi$. (3)
- 9 Give example of (a) removable singularity (b) pole (c) essential singularity (3)
- 10 Find the Laurent series expansions of $\frac{1}{z(z-1)}$ about $z = 0$ (3)

PART B*Answer any one full question from each module. Each question carries 14 marks***Module 1**

- 11 (a) Solve $y^2p - xyq = xz$ 7
- (b) Solve by the method of separation of variables $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$, $u(x, 0) = 4e^{-3x}$ 7
- 12 (a) Find the complete integral of $px + qy = pq$ using Charpit's method. 7
- (b) Form the partial differential equation corresponding to family of spheres with center on z -axis and radius a . 7

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Module 2

- 13 (a) Solve the boundary value problem described by $u_{tt}-c^2u_{xx}=0$, $0 \leq x \leq \ell$, $t \geq 0$ 7
 $u(0, t) = u(\ell, t) = 0$, $t \geq 0$, $u(x, 0) = 10 \sin\left(\frac{\pi x}{\ell}\right)$, $\frac{\partial u}{\partial t}(x, 0) = 0$
- (b) Find the temperature $u(x, t)$ in a homogenous bar heat conducting material of length l whose ends kept at 0°C and whose initial temperature is given by $u(x, 0) = lx - x^2$. 7
- 14 (a) Derive one dimensional wave equation. 7
- (b) The ends A and B of a rod 10 cm in length are kept at temperatures 0°C and 100°C until the steady state condition prevails. If B is Suddenly reduced to 0°C and kept so . Find the temperature distribution in the rod at time t . 7

Module 3

- 15 (a) Show that an analytic function $f(z) = u + iv$ is constant if its modulus is constant. 7
- (b) Find the image of $1 \leq |z| \leq 2$, $\frac{\pi}{6} \leq \theta \leq \frac{\pi}{3}$ under the mapping $w = z^2$ 7
- 16 (a) Verify whether $u = x^3 - 3xy^2$ is harmonic and find its conjugate harmonic function v . 7
- (b) Find the image of the region between real axis and a line parallel to real axis at $y = \frac{\pi}{2}$ under the mapping $W = e^z$ 7

Module 4

- 17 (a) Evaluate $\int_C |z|^2 dz$ where C is the circle $|z| = 2$. 7
- (b) Evaluate $\int_C \frac{z^2+2}{(z-3)^2} dz$ where C is the circle $|z| = 4$ using Cauchy's integral formula 7
- 18 (a) Evaluate $\oint_C \frac{e^z}{(z-1)(z-4)} dz$, where C is $|z| = 2$ using Cauchy's integral formula 7
- (b) Evaluate $\int \frac{3z^2+7z}{z+1} dz$ over (a) $|z| = 1.5$ (b) $|z + i| = 1$ 7

Module 5

- 19 (a) Find the Laurent series expansion for $f(z) = \frac{1}{(z-1)(z-2)}$ valid in (a) $1 < |z| < 2$ (b) $|z| > 2$ 7

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- (b) Evaluate $\int_0^{2\pi} \frac{d\theta}{5-4\sin\theta}$
- 20 (a) Evaluate $\int_{-\infty}^{\infty} \frac{x^2+2}{(x^2+1)(x^2+4)} dx$ 7
- (b) Using residue theorem evaluate $\oint_c \frac{z+1}{z^4-2z^3} dz$, where c is $|z| = \frac{1}{2}$ 7
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